



C1 · ADVANCED · ÜNİTE 7

Music, Memory & Emotion

Gramer: Complex Noun Phrase Post-modification (V-ing / V3 / to-inf / that-clause)

Ad Soyad: _____

Sınıf / Tarih: _____

Puan: _____ / 15

Müzik ve hafıza temasıyla isimleri V-ing, V3, to + V1 ve that-clause ile sonradan niteleyerek akademik isim öbekleri kurmayı öğreniyorsun.

1 · Kelimeler

2 · Gramer

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4 · Alıştırmalar

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Cevap anahtarı

1 Vocabulary · Kelimeler

#	WORD / CHUNK	TÜRKÇE AÇIKLAMA
1	involuntary memory	İstenç dışı bellek — Proustian effect
2	evocative	Çağrıştırmacı — güçlü duygu/anı uyandıran
3	nostalgia	Nostalji — geçmişe özlem
4	temporal lobe	Temporal lob — beynin bellek ve işitme merkezi
5	auditory cortex	İşitsel korteks — sesleri işleyen beyin bölgesi
6	procedural memory	Prosedürel bellek — kas hafızası, bisiklet sürmek gibi
7	semantic memory	Semantik bellek — olgular, bilgiler
8	episodic memory	Epizodik bellek — kişisel anılar
9	synesthesia	Sinestezi — duyarların karışması (renkleri duymak gibi)
10	neuroplasticity	Nöroplastisite — beynin kendini yeniden şekillendirme kapasitesi

2 Grammar · Gramer

Karmaşık isim öbekleri: isimden sonra gelen niteleyiciler

Türkçe'de niteleyici ismin **önüne** gelir (*müzik tarafından tetiklenen anılar*). İngilizce'de uzun niteleyiciler ismin **arkasına** gelir: *memories triggered by music*. Bunlar aslında kısaltılmış relative clause'lardır: *the woman (who is) sitting by the window*. Akademik metinlerde bu yapı bilgiyi özlü biçimde paketlemenin temel yoludur.

Özet tablo

Yapı / Kural	Form	Anlam	Örnek
Noun + V-ing phrase	Noun + V-ing = active post-modifier (doer of action)	İsimden SONRA gelen V-ing = aktif niteleme	<i>The researcher WORKING on memory... / A sound TRIGGERING involuntary memory...</i>
Noun + V3 phrase	Noun + V3 = passive post-modifier (receives action)	İsimden SONRA gelen V3 = pasif niteleme	<i>The memories STORED in music... / Evidence GATHERED by the team...</i>
Noun + to-infinitive	Noun + to + V1 = purpose/potential/following meaning	Amaç veya potansiyel — sıfat gibi	<i>The ability TO recall names / An opportunity TO extend the study / A tendency TO forget...</i>
Noun + that-clause	Noun + that + clause = content clause (abstract nouns)	Fikir, gerçek, sonuç isimleri sonrası	<i>The fact THAT music bypasses dementia / The claim THAT AI will replace musicians / The evidence THAT...</i>



⚠️ Post-modification = isimden SONRA gelen niteleme. Pre-modification = isimden ÖNCE. 'A TRIGGERING sound' (pre) vs 'a sound TRIGGERING memories' (post — longer, more complex). | Common abstract nouns + that-clause: fact, claim, theory, evidence, idea, possibility, belief, suggestion.

🔍 Örneklerle

✂️ Kısaltılmış relative clause

People **living** near the coast... → = *people who live near the coast* (kıyıda yaşayan insanlar)

The songs **chosen** for the study... → = *the songs that were chosen* (çalışma için seçilen şarkılar)

🎯 İsim + to-infinitive / edat öbeği

the **first person to reach** the summit → *zirveye ulaşan ilk kişi*

a **decision to leave** the company → *şirketten ayrılma kararı*

a book **about the history of jazz** → *caz tarihiyle ilgili bir kitap*

💡 Soyut isim + that-clause

the **fact that** music can reduce stress →
müziğin stresi azaltabildiği gerçeği

the **idea that** memory is fixed → belleğin sabit
olduğu fikri

There is **growing evidence that** sleep
improves learning. → Uykunun öğrenmeyi
geliştirdiğine dair artan kanıtlar var.



IMPORTANT — that-clause ≠ relative clause

the fact **that** he lied → that gerçeğin **içeriğini** verir (atılamaz, which olamaz).

the story **that** he told → relative clause (which olabilir, nesne olduğu için atılabilir).



COMMON MISTAKE — V-ing ile V3'ü karıştırmak

~~the songs choosing for the study~~ → **the songs chosen** for the study (şarkılar seçmez,
seçilir → V3)

~~people lived near the coast~~ → **people living** near the coast (insanlar yaşar → V-ing)



Kendine sor — pratik yap

The number of people _____ (use) smartphones has doubled. (etken)

Most of the goods _____ (sell) online are cheaper. (edilgen)

She was the first woman _____ (win) the prize.

There's no evidence _____ the product works.

3 Reading · Okuma

The Proustian Effect

A piece of music heard in childhood can activate memories **that had seemed permanently lost**. This phenomenon — the ability **of sound to reach** into the neural architecture of the past — has become one of the most studied and most debated areas in cognitive neuroscience.

Patients **suffering from advanced Alzheimer's disease**, their verbal communication almost entirely gone, have been observed responding to music **stored in what researchers call procedural and emotional memory**. The pathways

used to access this music appear largely unaffected by the disease — offering a remarkable window into the mystery of identity.

The fact that music can access neural pathways largely untouched by dementia is extraordinary. Researchers working in this field — neurologists, psychologists, and music therapists — speak of "involuntary memory": the experience of being suddenly transported, without warning or will, by a sound carrying emotional information laid down decades earlier.

The claim that music could serve as a therapeutic tool for dementia patients is now widely supported by evidence gathered in dozens of clinical settings. The desire to harness this effect therapeutically has driven significant research investment.

Music, it seems, speaks to a part of the self that language and conscious memory cannot reach. The question occupying researchers today is not merely whether this is true, but how — and the possibility that the answer lies in the deepest structures of human identity is one few scientists are comfortable dismissing.

İ Gerçek olay ve araştırmalara dayanır; ders amacıyla sadeleştirilmiştir.

Kaynaklar:

• Jacobsen vd. (2015), Brain — Why musical memory can be preserved in Alzheimer's disease — <https://academic.oup.com/brain/article/138/8/2438/330051>

Comprehension Questions

1. What is 'involuntary memory' according to the text?

2. What is remarkable about music and Alzheimer's patients?

3. Find 3 examples of noun + V-ing post-modification in the text.

4. Find 2 examples of noun + that-clause in the text and explain what type of abstract noun they follow.

4 Exercises · Alıştırmalar

Doğru seçeneği işaretleyin.

- | | |
|---|--|
| <p>1. The researcher ___ in this field for the past decade has published groundbreaking work.
(a) worked (b) to work (c) working</p> <p>2. The memories ___ in music are often the very last to fade.
(a) storing (b) to store (c) stored</p> <p>3. She has the ability ___ complex emotional subtext in her performances.
(a) conveying (b) conveyed (c) to convey</p> | <p>4. The fact ___ music bypasses damaged memory pathways is clinically significant.
(a) which (b) of (c) that</p> <p>5. The patients ___ in the clinical study showed remarkable and unexpected improvement.
(a) participated (b) to participate (c) participating</p> <p>6. The evidence ___ by the research team strongly confirms the initial hypothesis.
(a) gathering (b) gathered (c) to gather</p> |
|---|--|

7. The opportunity ___ the study into a clinical trial may not arise again.
(a) extending (b) extended (c) to extend
8. The questions ___ by this research touch on the very nature of human identity.
(a) raising (b) to raise (c) raised
9. The phenomenon ___ the scientific community for decades is finally yielding to explanation.
(a) puzzled (b) to puzzle (c) puzzling
10. The claim ___ music can reverse memory decline entirely remains controversial.
(a) which (b) that (c) of
11. The theory ___ by this research group has gained international recognition.
(a) proposing (b) to propose (c) proposed
12. The desire ___ these findings to therapeutic practice is driving further research.
(a) translating (b) translated (c) to translate
13. The music ___ in the temporal lobe appears to activate deep emotional networks.
(a) processing (b) processed (c) to process
14. Patients ___ from severe amnesia often retain their musical memory intact.
(a) suffered (b) to suffer (c) suffering
15. The conclusion ___ the team reached after three years of study surprised even its members.
(a) which (b) of (c) that

5 Writing · Yazma

1. Post-modification alıştırması

Write 5 complex noun phrases with post-modification:

2 with V-ing (active)

2 with V3 (passive)

1 with that-clause

Topic: memory, music, or the brain.

2. Müzik ve bellek

Write a paragraph (150 words) about music and memory using complex noun phrases with post-modification throughout. Try to avoid simple adjectives — use participial and clausal post-modifiers instead.



Answer Key · Cevap Anahtarı

Exercises

- 1. c) working** — Noun + V-ing phrase (active post-mod): the researcher working = who works
- 2. c) stored** — Noun + V3 phrase (passive post-mod): memories stored = that are stored
- 3. c) to convey** — Noun + to-infinitive: the ability to convey (potential/capacity)
- 4. c) that** — Abstract noun (fact) + that-clause = content clause
- 5. c) participating** — Noun + V-ing (active): patients participating = who participated
- 6. b) gathered** — Noun + V3 (passive): evidence gathered = that was gathered
- 7. c) to extend** — Noun + to-infinitive: the opportunity to extend (future possibility)
- 8. c) raised** — Noun + V3 (passive): questions raised = that are raised
- 9. c) puzzling** — Noun + V-ing (active): phenomenon puzzling = that puzzles
- 10. b) that** — Abstract noun (claim) + that-clause = content clause (contested assertion)
- 11. c) proposed** — Noun + V3 (passive): theory proposed = that was proposed
- 12. c) to translate** — Noun + to-infinitive: the desire to translate (purpose/goal)
- 13. b) processed** — Noun + V3 (passive): music processed = that is processed
- 14. c) suffering** — Noun + V-ing (active): patients suffering = who suffer
- 15. c) that** — Noun (conclusion) + that-clause = appositive content clause

Comprehension

1. Being suddenly transported, without warning or will, by a sound carrying emotional information from decades earlier — the Proustian effect.
2. The neural pathways used to access music appear largely unaffected by the disease, even when verbal communication is gone.
3. e.g. 'Patients SUFFERING from Alzheimer's' / 'Researchers WORKING in this field' / 'The question OCCUPYING researchers today' / 'a sound CARRYING emotional information'
4. e.g. 'The fact THAT music can access neural pathways...' (fact = established truth) / 'The claim THAT music could serve as a therapeutic tool' (claim = contested assertion)

Grammar practice

1. The number of people ____ (use) smartphones has doubled. (etken) → **using**
2. Most of the goods ____ (sell) online are cheaper. (edilgen) → **sold**
3. She was the first woman ____ (win) the prize. → **to win**
4. There's no evidence ____ the product works. → **that**

Writing — örnek cevaplar

Görev 1 — Post-modification alıştırması

The students suffering from exam anxiety often struggle to recall information they know well. Researchers working in this field have made significant recent breakthroughs. The claim that music can access neural pathways unaffected by dementia has changed how we treat Alzheimer's patients. The question occupying neuroscientists today is why music memory seems to be encoded differently.

Görev 2 — Müzik ve bellek

Music is a phenomenon occupying a unique place in human experience. The neural pathways activated by a familiar song are pathways untouched by many neurological conditions. The claim that music can restore identity — even briefly — in patients suffering from severe memory loss is a claim supported by growing evidence. What remains to be understood is the mechanism underlying this extraordinary effect.

İnteraktif sürüm, anında geri bildirim ve XP için: abonlinenglish.school/ingilizce-c1/unite-7-noun-phrase-post-modification